

Cellular plasticity: 1712 to the present day

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Cell identity is a fundamental feature of cells. Tissues are often organized into cellular hierarchies characterized by progressive differentiation and developmental commitment. However, it is been historically evident that the cells of many organisms of various phyla, especially in the context of injury, exhibit remarkable plasticity in terms of their ability to convert into other cell types. Recent modern studies, using genetic lineage tracing, have demonstrated that many mature functional cells retain a potential to undergo lineage reversion (dedifferentiation) or to convert into cells of other more distant lineages (transdifferentiation) following injury. Similarly, mimicking progenitor cell transdetermination, stem cells can interconvert. These forms of plasticity may be essential for organismal survival, and are likely part and parcel of regeneration.

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Introduction to plasticity

Multicellular organisms often need to maintain their form and function by continuously generating new cells to replace older cells that have been lost in the process of normal wear and tear. When the equilibrium of new cell generation and steady state cell loss is perturbed by tissue injury, homeostatic mechanisms are invoked to allow regeneration of damaged tissue. Until recently, it was thought that this equilibrium was, in the main, restored through the replication of adult stem/progenitor cells and their subsequent differentiation or through the replication of mature differentiated cells. These homeostatic cellular

mechanisms were thought to obey defined lineage hierarchies, but it is becoming increasingly clear that classical directional lineage hierarchies do not define all the physiologically relevant paths a regenerating cell can tread.

During development, from egg to embryo, embryonic progenitor cells differentiate into progressively more diverse cell types. These events are thought to occur in such a way that several distinct cell intermediates are generated, with increasingly restricted lineage potential, until the final mature specialized cell types are generated and functionally integrated into their respective tissues. This general schema has been indelibly imprinted in our thinking by Konrad Waddington through his use of cartoons to depict the so-called epigenetic landscape of the embryo [1]. An implicit corollary to these notions is that progressively mature cells irretrievably lose the potential to give rise to progeny outside of their given lineage. That said, much earlier in the history of embryology, as far back as the late 1800s, August Weismann's and Wilhelm Roux's notion that embryonic cell fate was 'determined' with each subsequent cell division of the embryo, stood in contrast to the results of Han Driesch's experiments that suggested that early embryonic cells were plastic or 'regulative' and could respond to external injury [2]. More specifically, when Roux used thermal injury to kill one of the cells of a 2 cell frog embryo, the resulting larva possessed only a right or left half, suggesting that even early embryonic cells were 'determined' [2]. In contrast, Driesch's isolation of a single blastomere from an early multicellular sea urchin embryo, suggested that a single isolated blastomere could produce an entire larva, suggesting that sea urchins possessed 'regulative' development where multiple embryonic cells retain a potency to form an entire organism [2].

Harkening back to these very early seemingly discrepant findings, later studies challenged the notion that adult differentiated cells are irreversibly committed to a particular fate, both in experimentally-induced and physiological conditions. In a remarkable example of experimentally induced reprogramming, Briggs and King in 1952 managed to generate frog tadpoles by transplanting the nuclei of cells from the blastula into *Xenopus* oocytes [3]. John Gurdon then showed that this reprogramming could be accomplished with even more differentiated cells [4–6] and this body of work eventually culminated with the cloning of a mammal [7]. Less well known work from the laboratory of Ernest Hadorn revealed that fly imaginal disc progenitors from one imaginal disc could 'transdetermine' and acquire the characteristics of different imaginal disc progenitor cells when transplanted from one larva to a

heterologous site in a second larva (Figure 1a). In 1987, it was then shown that ectopic expression of the *Antennapedia* homeotic gene led to changes in the body plan of flies, such that leg appendages appeared where antennae should have formed [8]. Similarly, studies revealed that ectopic expression of the *eyeless* gene could lead to the formation of ectopic eyes where normal legs should have formed [9]. Subsequently, the remarkable capacity of *MyoD* to reprogram disparate cells into muscle cells set the stage for modern iPSC and direct cell reprogramming strategies, therein completing an arc of experiments concerned with ‘artificially’ induced cell plasticity [10,11]. Herein we would like to give an overview of the historical and modern experimental basis for thinking about cell plasticity as a normal physiologic agency following injury-induced regeneration. Stated otherwise, we endeavor to show that cell plasticity is not ‘unnatural’.

Historical perspectives on adult cell plasticity in regeneration

Some of the first descriptions of regeneration date back to 1712, when Swiss scientist Abraham Trembley noted that the freshwater polyp *hydra* regenerates after being cut in half. In his descriptions from his treatise ‘*Mémoires, Pour Servir à l’Histoire d’un Genre de Polypes d’Eau Douce, à Bras en Forme de Cornes*’, he noted that when polyps were cut into two vertical halves, each part gave rise to two smaller, but fully intact, normally re-patterned organisms [12]. In 1769, Spallanzani described how tadpoles could regenerate their tails and how salamanders could regrow amputated limbs, tails and jaws (*An assay on animal reproductions, Spallanzani, 1769*) [13]. In 1895, Wolff used a model of lens extirpation in the newt to show that missing lens tissue was, in fact, regenerated from developmentally distinct iris pigment epithelial cells. The pigment cells first dedifferentiated into non-descript cells without pigment, and then transdifferentiated into lens cells (Figure 1b) [14,15]. Thus, tissue-level observations from long ago set the stage for our current modern exploration of cell plasticity [13].

Terminology and definitions

Cellular plasticity during regeneration has now been scrutinized in many model organisms and tissues using inducible cell type-specific lineage tracing and in some cases by direct visualization. In this review, we refer to dedifferentiation as a process of lineage reversion in which differentiated cells acquire the properties of more immature cells within the same lineage hierarchy. Transdetermination classically refers to the conversion of one progenitor/stem cell population into another, thereby potentially forming a basis for a metaplastic tissue transformation [16–19]. Transdifferentiation, in contrast, refers to the conversion of one differentiated cell type into another, thereby affording another possible mechanism for tissue level metaplasia. Indeed, all 3 of these

processes may occur in different contexts, and with further studies we may find that they all occur within the same tissues in various differing degrees based upon the extent and specific nature of the tissue injury. We would like suggest that each case of injury in each tissue must be examined individually before any general conclusions about the nature of plasticity can be drawn. Below we present an incomplete synopsis of some of the first such experiments that may be illustrative of more general principles of plasticity, focusing on vertebrate cell plasticity.

Cellular plasticity in invertebrates

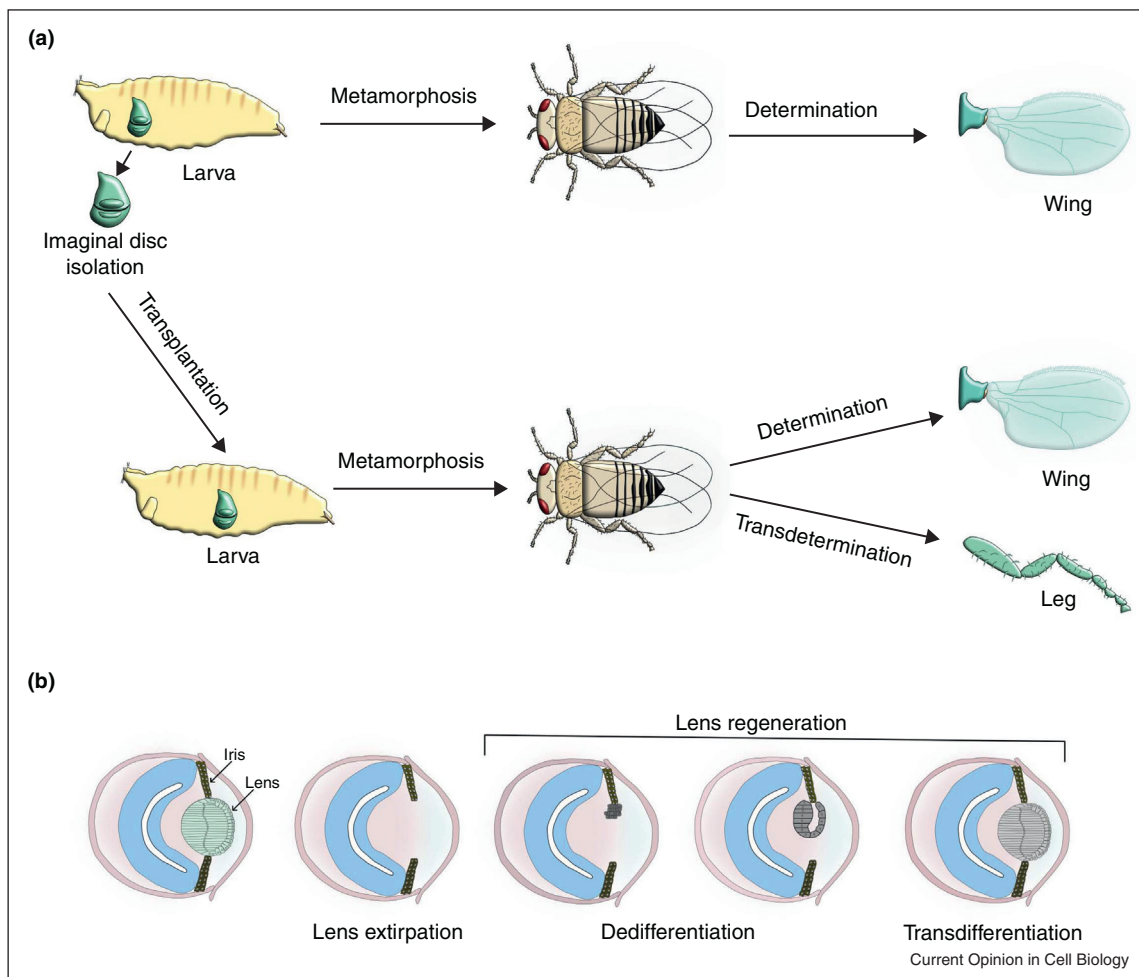
In the fly gerarium, when female or male germ stem cells are lost either via genetically forced differentiation or by laser ablation, differentiating gonialblasts and spermatogonia were shown to restore the missing germ cell population via dedifferentiation [20]. In male flies, interconnected spermatogonia lose their ring canals and separate into single cells to form functional germ stem cells [21,22,23]. Similar plasticity has been noted in the mouse testis where interconnected spermatogonial clusters fragment into single cells and act as stem cells [24,25]. Of note, in both male and female fly germaria, no terminally differentiated oocytes or spermatocytes have been shown to possess the capacity to dedifferentiate into stem cells. The remarkable regenerative capacity of planarians may further teach us about fundamental mechanisms of cell plasticity, but it remains unclear how insights into the biology of neoblasts will apply to vertebrates [26].

Cellular plasticity in vertebrates

When newt limb is amputated, a cluster of seemingly dedifferentiated progenitor cells, referred to as the blastema, appear and these cells then give rise to the tissues of the newly regenerated limb. Interestingly, even in the case of newt limb regeneration, it appears that the axolotl regenerates differentiated limb cells in a lineage-restricted form of dedifferentiation [27]. Remarkably in another newt, *Notophthalmus viridescens*, following limb amputation Pax7+ stem cells regenerate myocytes without large scale dedifferentiation. Thus, in seemingly closely related organisms, dedifferentiation and more conventional stem cell differentiation can be deployed to differing degrees to effect a superficially similar form of regeneration [28,29]. This again points to the need to study cell plasticity within particular defined contexts, and the need to be cautious about generalizing experimental findings.

Zebrafish can regenerate their hearts following partial amputation. During this process, cardiomyocytes dedifferentiate and proliferate to regenerate missing ventricular tissue. Specifically, the sarcomere contractile apparatus and myosin heavy chain proteins are lost prior to myocyte replication, indicating that these dedifferentiated cells have undergone a complex structural, molecular and morphological change during this process (Figure 2a)

Figure 1



Historical examples of cellular plasticity. **(a)** Transdetermination following fly imaginal disc transplantation. **(b)** Dedifferentiation followed by transdifferentiation regenerates a complete new eye following lens extirpation.

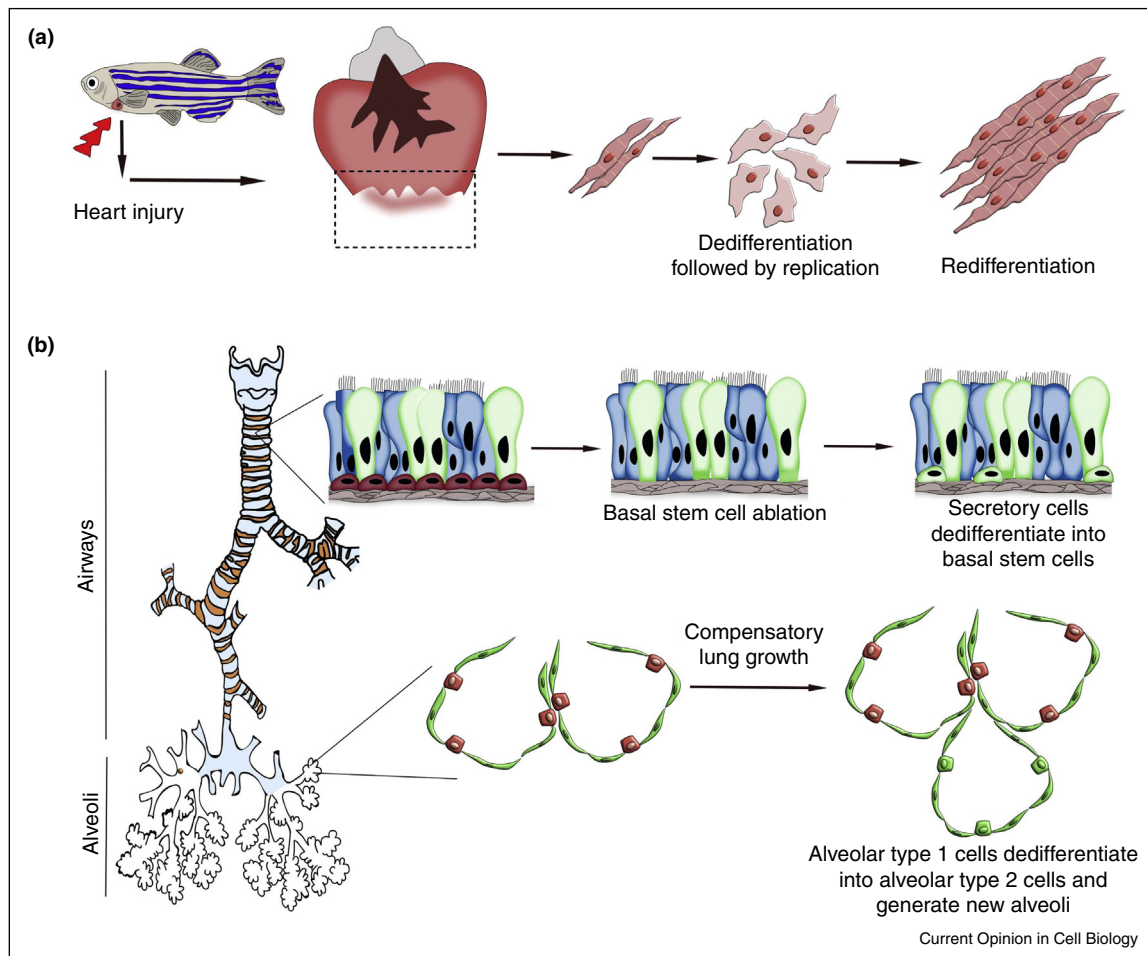
[30,31^{••},32]. In the newt limb and in the zebrafish heart, lineage relationships seem to be maintained and dedifferentiated progenitors are only present in any great number during regeneration [27,29[•],31^{••},32]. Of note, this transient appearance of progenitor cells is also a feature of the lenticular transdifferentiation noted above, but as we will see is not a feature of all forms of cell plasticity.

Cell plasticity in mammals

In mammals, intestinal crypt Dll1+ proliferating secretory progenitors cells have been shown to acquire Lgr5+ stem cell characteristics when these stem cells are damaged by irradiation [33^{••},34[•]]. Intriguingly, seemingly terminally differentiated, digestive enzyme-producing Troy+ chief cells located in the base of gastric glands actually give rise to all of the cell types of the gastric glands at low rates during normal tissue homeostasis [35^{••}]. However after damage, induced by cell ablation, this tendency for 'reversion' is enhanced [35^{••}]. It remains possible that some

special subpopulation of Troy+ cells normally turnover slowly to give rise to mature Chief cells, but that these same progenitor cells exhibit plasticity after injury. In airway epithelium, fully mature functional secretory cells have been shown to stably dedifferentiate into *bona fide* basal stem cells following similar stem cell ablation (Figure 2b) [36^{••}]. Depending on the injury, in this epithelium, regeneration can occur through conventional stem cell differentiation or in the absence of stem cells, secretory cells can act as facultative stem cells by using dedifferentiation to repopulate missing stem cells. This causes one to wonder whether the axolotl and *Notophthalmus* both use a combination of stem cell differentiation and dedifferentiation to generate new myocytes, but exploit these processes to differing degrees, perhaps based on the nature of the injury to the limb. In addition, in all these cases, in contrast to the amphibian experiments described above, clonal dedifferentiated cells have been shown to persist long term and participate in the

Figure 2



Dedifferentiation. **(a)** Transient dedifferentiation and redifferentiation of zebrafish cardiomyocytes following ventricular amputation. **(b)** Stable dedifferentiation of mature secretory cells of the airway epithelium into long lived basal stem cells (upper panel) and dedifferentiation of AEC type 1 cells into AEC type 2 cells following partial pneumonectomy.

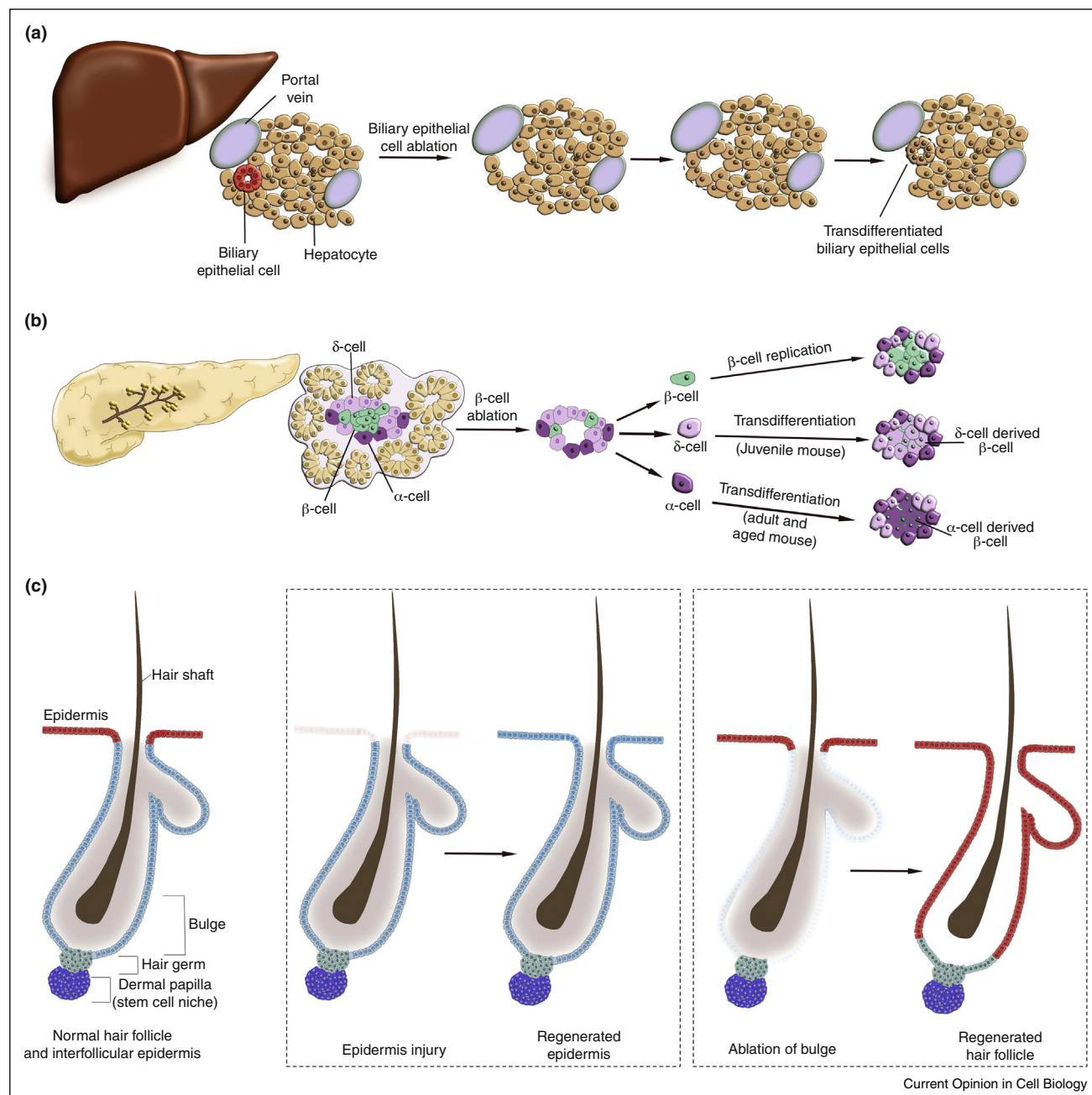
post-regeneration steady state turnover of the respective epithelia [36^{••}]. In alveoli, type 2 cells normally serve as stem cells, and differentiate into type 1 cells [37]. However, recent lineage-tracing studies indicate that differentiated type 1 cells can lose their remarkably complex morphology and generate type 2 stem cells. These dedifferentiated type 1-derived stem cells can then generate new type 1 and type 2 cell-containing alveoli following pneumonectomy-induced lung regeneration (Figure 2b) [38[•]].

In the liver, a paradigm of mammalian regeneration, following biliary cell toxic injury, hepatocytes can transdifferentiate and form bile duct epithelial cells (Figure 3a) [39[•]]. After kidney injury, differentiated cells of the proximal convoluted tubule epithelium have been shown to transiently express embryonic kidney markers, as they replicate to restore lost epithelial cells [40[•],41]. In the pancreas, ablation of β -cells *in vivo* induces β -cell

replication to restore the missing population. However, it has also been documented that pancreatic α -cells and δ -cells can transdifferentiate to replace lost insulin-producing β -cells [42,43[•]]. Remarkably, in the adult and aged mice, pancreatic α -cells are the source of transdifferentiation while in the juvenile mouse, δ -cells subtend this regenerative response (Figure 3b). In the intestine, Hopx/Bmi1/Tert⁺ cells can serve as reserve stem cells, and can convert into Lgr5⁺ cells to replace ablated Lgr5⁺ cells [33^{••},34[•],44–48]. Thus, in these cases, stem cells can convert into other types of stem cells. Such interconversions of diverse, but related populations of stem cells, closely parallel the transdetermination events that Hadorn described in transplanted fly imaginal disc progenitors [16,18].

In mammary tissue, lineage tracing studies have demonstrated that distinct stem cells maintain basal and luminal cell compartments in the steady state epithelium [49–52].

Figure 3



Transdifferentiation and transdetermination. **(a)** Transdifferentiation of hepatocytes into biliary duct epithelial cells following toxin-induced biliary injury. **(b)** Replication of β -cells, or transdifferentiation of α cells and δ -cells into β -cells in response to β -cell injury. **(c)** Stem cell plasticity following epidermal and hair follicle injury.

However, recent studies have suggested that rare basal cells can generate both basal and luminal cells of the mammary epithelium during both homeostatic maintenance and during lactation [50,52]. The role of *in vivo* plasticity in this system warrants further scrutiny since near-saturation lineage tracing coupled with mathematical modeling suggests that unipotency is the dominant process

in mammary gland postnatal homeostasis and pregnancy and lactation [53]. Intriguingly, in a different context, when unipotent basal cells are transplanted into another mouse's mammary fat pad, they have the potential to form both luminal and basal cells, demonstrating that basal cells possess an unexpected potency and plasticity revealed by experimental perturbation, just as the blastomere isolation

experiments of Driesch suggested in the ‘regulative’ sea urchin embryo [2,49]. Indeed, in the disease context, oncogene expression in unipotent mammary gland progenitors elicits the production of pathologic multipotent progenitors [54,55[•]]. Studies on prostate epithelium reveal the presence of basal progenitors, luminal progenitors, and rare multipotent stem cells [56–62]. Recent studies have demonstrated that human prostate luminal cells can serve as multipotent progenitors in transplantation assays [61,63,64]. Indeed, these studies in the mammary and prostate gland, point to the notion that the clonal tracing of cells, each with their own unique genetic markers, as well as saturation labeling will eventually be required in all model systems if we are to truly pin down the nature of cellular plasticity, since plasticity, at the end of the day, is a process that involves a single cell converting into another. Such concerns indeed raise the notion of how to define unique cell type identity in the age of single cell analysis at the molecular level.

Other archetypal examples of plasticity occur in the adult skin which harbors several different stem cell populations in distinct tissue domains [62,65–70]. Lineage tracing analysis reveals that these stem cells strictly maintain their separate compartments during steady state epithelial maintenance (Figure 3c). However, after severe injury, stem cells in one compartment are able to generate almost all the differentiated cell types of other compartments, as well as their respective stem cell populations [71[•],72–75]. This again, as in the case of intestinal stem cell interconversion, is analogous to the transdetermination events seen after fly imaginal disc transplantation [16,18]. Specifically, hair follicle stem cells have been shown to contribute to the interfollicular epidermis after severe injury to the latter and/or following transplantation [72–74,76–78] (Figure 3c). Conversely, when the hair bulge was ablated, stem cells of the interfollicular epidermis and/or hair germ cells repopulated the lost bulge cell population (Figure 3c). Whether these cells remain stable over time has yet to be assessed. Additionally, hair germ cells were shown to recolonize bulge stem cells after their ablation. Using live imaging coupled with photoablation, this study elegantly demonstrated that hair germ cells could convert to bulge stem cells and then participate in hair growth [71[•],79]. However, again, long-term lineage tracing needs to be performed to understand whether these new bulge cells participate in recurring hair cycles. In aggregate, these findings all point to a remarkable cellular plasticity that allows tissue reconstitution after injury, whether this be in the form of transdifferentiation, dedifferentiation, or the conversion of one stem cell population into another. Further attesting to how different forms of tissue regeneration may occur through differing cellular mechanisms despite a seemingly similar phenomenology, recent studies in two species of African spiny mouse, *Acomys kempfi* and *Acomys percivali*, demonstrate the appearance of a blastema-like structure.

Indeed, this may explain why these animals can regenerate whole swaths of skin with intact hair follicles and ear cartilage, in contrast to other species of mice [80].

Restraining plasticity: How do cells maintain their state?

In the case of airway epithelium, lineage tracing has revealed that dedifferentiated basal stem cells can persist long term and participate not only in the steady state maintenance of the epithelium, but also to physiologic injury repair [36^{••}]. In these studies, the capacity of differentiated cells to undergo dedifferentiation into a stem cell was inversely proportional to their degree of maturation. Interestingly, plasticity itself is generally more evident in young organisms than their aging counterparts [81]. The mechanisms dictating this loss of plasticity with age or differentiation should be a topic of great interest in the coming few years. Insights into such mechanisms could provide us with tools for enhancing regeneration, but also likely will inform our understanding of the pathologic plasticity associated with cancer.

Generally speaking, plasticity is fundamentally regulated as much by non-cell autonomous mechanisms as cell autonomous ones. For example, stem cells are often located in a defined microenvironment, referred to as a niche. This microenvironment, by definition, is necessary for stem cell maintenance. Historically, the term niche was first postulated by Robert Schofield in 1978 in his studies of hematopoietic stem cells and their relationship to the bone marrow microenvironment [82]. Now it is clear that most every tissue-resident stem cell is influenced by its surrounding microenvironment, and in some cases that microenvironment has been shown to subtend a niche function, in that elements of the microenvironment are actually necessary for the maintenance of the stem cell population being considered [65,83]. Recent studies have refined our understanding of the mechanisms of niche action. In the drosophila testis, it has been noted that hub cells, which act as niche cells for germ stem cells, are necessary for the dedifferentiation of differentiating spermatogonial cells into germ stem cells [22]. Intriguingly, the distance between hub cells and the dedifferentiating cells determines the propensity for cellular dedifferentiation [22,23,84]. This phenomenon is based upon a gradient of a short-range morphogen from the hub cell which is, at least in part, the molecular basis for the hub cell niche function [23,84–87]. Similarly, in hair follicles, the ability of regenerating lost bulge stem cells is dependent upon signals derived from the dermal papillae [65,71[•],79,83]. In the airway epithelium, the ability of mature secretory cells to dedifferentiate is actually restrained in the presence of the actual stem cells themselves [36^{••},88^{••}]. Using *ex vivo* organoid culture, clonally cultured secretory cells dedifferentiated into stem cells. However, dedifferentiation was completely inhibited when differentiated secretory cells were cultured in close proximity to stem cells [36^{••}].

Thus, basal stem cells themselves actually suppress the ability of secretory cells to dedifferentiate through what is likely a contact mediated form of cell-cell communication [36^{••},88^{••}]. Interestingly, basal stem cells continuously supply Notch ligands to their own secretory cell progeny, and these stem cell-derived ligands serve as progenitor cell niche signals in that they are necessary for the continued maintenance of the daughter secretory cells. Indeed, Notch ligands can be therapeutically blocked to engender cell fate transformations in the context of allergic disease [89]. In the absence of the basal cell ligands, the secretory cell daughters differentiate. Such *feed-forward* signals thus constitute a stem cell-derived niche signal for daughter secretory cells in the same manner that stem cells themselves require a microenvironment to be maintained within a tissue [88^{••}]. Other studies in skin have demonstrated that tissue wide signals, in analogy to bacterial quorum sensing, are used for the coordination of stem cells in neighboring hair follicles to ensure coordinated tissue-level regeneration [90]. Thus, in addition to the cell autonomous factors known to regulate the stability of cell fate, plasticity is also restrained *in vivo* by short-range signals, long-range signals, or contact-mediated communication with neighboring microenvironments or their physical properties.

Conclusions

Lineage tracing and live imaging studies have greatly improved our understanding of how tissues maintain appropriate cell types and numbers during homeostasis, and how they respond to different injury stimuli. Recent studies have made attempts to define the contributions of cellular plasticity in the form of dedifferentiation, transdifferentiation, and transdetermination *in vivo*. Indeed plasticity likely serves a physiologic function in the setting of myriad tissue injuries such as infection, allergen response, or toxic injury. Unfortunately, unwanted plasticity likely contributes to pathology in disease states where the normal regenerative response is circumvented, such as in cancer.

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